Gauraang Varshney

COMPUTATIONAL ASTROPHYSICS · STAR & PLANET FORMATION

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Research Interests

Computational astrophysics and numerical modelling of planet and star formation; dust coagulation, turbulence and transport in protoplanetary disks; hydrodynamical simulations and cosmic dust; with a particular interest in messy and chaotic problems, using theoretical models and computation to uncover the elegant structure beneath them.

Education

Master of Science (M.Sc) in Physics — India
Indian Institute of Technology, Roorkee (IIT Roorkee) — July 2025 – 2027
GPA: **8.78/10**
Relevant Coursework: Nuclear Astrophysics, Statistical Mechanics, Advanced Atmospheric Physics, Computational Physics, Programming in C/C++

Bachelor of Science (B.Sc) in Physics (Honours) — India
Gautam Buddha University (GBU) — 2022 – 2025
CGPA: **9.66/10, Department Topper**
Relevant Coursework: Introductory Astrophysics (Grade A), Electromagnetic Theory, Classical Mechanics, Quantum Mechanics

International Baccalaureate (IB) Diploma — Namibia
Windhoek International School — 2020 – 2022
7/7 in HL Physics, HL Mathematics, and HL Chemistry. Extended Essay in Physics with grade A.

Research Experience

M.Sc Thesis: Dust Evolution and Turbulence in Protoplanetary Disks — Roorkee, India
Indian Institute of Technology, Roorkee — Jan 2026 – Ongoing
Supervised by Dr. Akshay Bhatnagar, Assistant Professor at IIT Roorkee.
Building a vertically resolved, multi-species dust-coagulation model in Python to study how grain growth competes with turbulent stirring across disk radii and heights. Derived analytic settling profiles and a seed-size-independent midplane grain size, validated the numerical solutions against them, and am now extending the code with Monte Carlo coagulation using turbulent relative velocities.

Astronomy Winter Internship: Differential Photometry of Variable Stars — Nainital, India
Aryabhatta Research Institute of Observational Sciences (ARIES) — Dec 2025 – Feb 2026
Under Dr. Yogesh Chandra Joshi, Scientist at ARIES.
Developed a differential photometry pipeline [On Github] for time-series analysis of variable stars using multi-epoch optical observations from the 3.6-m Devasthal Optical Telescope. Applied Lomb–Scargle and Box Least Squares (BLS) methods for periodicity detection and classified variable-star candidates through cross-matching with TESS and Gaia catalogues.

Summer Research Internship: Cosmological Parameters from Pantheon Type-Ia Remote
India Space Academy, Department of Space Education June 2025 – July 2025
Advisor: Satyapriya Das, ISRO Scientist
Analysed the Pantheon Type-Ia supernova dataset to study cosmological parameters and the age of the Universe using Python-based numerical methods. Performed model fitting and residual analysis to investigate the consistency of different cosmological models with the observational data. [On Github]

Bachelor Thesis: Simulating Quantum Algorithms for Cryptography
Gautam Buddha University Jan 2025 – May 2025
Supervisor: Dr. Venus Dillu, Assistant Professor at GBU.
Developed Python-based simulations of quantum computing algorithms for cryptographic applications using Qiskit [On Github] . Implemented data structures and parameter-optimization methods to study the behaviour of quantum circuits relevant to the project.

Independent Projects

Handheld Solar Irradiance & UV Index Detector Roorkee, India
Working project now used in the IITR Atmospheric Physics Lab Aug 2026 – Ongoing
Designing and building a portable solar radiation measurement instrument using an ESP32 microcontroller, LTR390 UV sensor, and AS7341 spectral sensor. Developing embedded firmware for real-time sensor acquisition and OLED display of UV index and optical measurements, while working on sensor calibration, irradiance estimation, circuit integration, and a custom handheld enclosure using CAD.

Course Project: Simulation-based Analysis of Gaussian Wavepackets
Gautam Buddha University Jan 2024
Developed a MATLAB GUI for numerical and visual analysis of Gaussian wavepackets. Simulated the time-dependent evolution and implemented interactive visualization of the wavepacket dynamics. [On Github]

Honours & Awards

- **2025** **International Astronomical Search Collaboration:** Recognition for contributions to the analysis of Pan-STARRS observations of near-Earth objects and Main Belt asteroids. – IASC, Cisco, NASA, Pan-STARRS.
- **2025** **Best Research Poster Award:** Awarded for outstanding research presentation at the National Science Day 2025 poster competition. – Gautam Buddha University.
- **2025** **QBronze Award:** Awarded upon completion of the Quantum Solver training programme on quantum algorithms and problem solving. – WISER & Womanium.
- **2024** **Merit Scholarship:** Awarded by Gautam Buddha University for academic performance in the department.
- **2024** **IUAC Summer Fellowship:** Awarded a fully funded fellowship for the Summer Training Programme at the Inter-University Accelerator Centre (IUAC), a national user facility housing India's largest tandem Pelletron accelerator.
- **2022** **Academic Excellence Award:** Awarded NAD$10,000 by Windhoek International School, Namibia.

Seminar Presentations & Posters

2026 **Seminar:** *Dust Evolution and Turbulence in Protoplanetary Disks*. Explored the physical mechanisms governing planetesimal formation and the conditions required for overcoming growth barriers in turbulent protoplanetary disks. – IIT Roorkee.

2025 **International Symposium on Quantum Information and Communication (ISQIC):** Poster on enhancing the BB84 quantum key distribution protocol using expanded polarization states, based on my B.Sc. thesis work. – TCG CREST & CQuERE, Kolkata.

2024 **Contributed Talk:** Presented the working principle and detection mechanism of spark counters for α-particle detection and range measurements in air. – IUAC, India.

2024 **National Science Day:** Poster on the Moon Impact Probe of Chandrayaan-1 and its role in India's first lunar mission. – Gautam Buddha University.

Technical Skills

Programming: Python *(NumPy, SciPy, Matplotlib, Pandas, Astropy, Lightkurve)*, C/C++ (basics), FORTRAN (basics), MATLAB, Scilab, LaTeX

Numerical Methods: ODE integration, PDE solvers, finite-difference methods, Monte Carlo methods, numerical simulation, model fitting, scientific visualization

Astronomical Data: FITS analysis with SAOImageDS9, photometry, Lomb–Scargle and BLS periodograms, TESS and Gaia DR3 catalogues

Additional Training

Introduction to Astrophysical Fluids — NPTEL
IIT Kanpur — 2026 – Ongoing
Instructor: Dr. Supratik Banerjee. Currently undertaking coursework on astrophysical fluid dynamics, with emphasis on the governing equations and their application to astrophysical systems and disks.

Analyzing the Universe — Coursera
Rutgers, The State University of New Jersey — Dec 2025
Instructor: Dr. Terry A. Matilsky. Training in FITS analysis, energy spectra, and astronomical time-series data using SAOImageDS9.

Summer Training Programme — Delhi, India
Inter-University Accelerator Centre (IUAC) — June 2024
Supervised by Dr. Sarvesh Kumar and Mr. S. K. Suman. Conducted an on-site experimental and reading project on α-particle detection and range measurements using a radioactive ^{241}Am source.

Extracurriculars

Outreach: **Explain to Us!** — Co-founder & Co-organizer: Co-founded and coordinate a student-led science outreach initiative that connects university students with researchers through accessible online talks and discussions. Lead international speaker outreach and event planning, building engagement with scientists across institutions in India and abroad.

Sports: **Swimming** – 3rd Place, Aquatic Meet IITR 2025; **Chess** – 1st Place, Inter-Year IITR 2025; 1st Place, Inter-Department Tournaments GBU 2024–25.

Positions of Responsibility: **Design Member**, International Relations Cell, IIT Roorkee; **Event Manager**, Thomso 2025, IIT Roorkee; **Head of Editorial**, C.V. Raman Club of Science and Technology, GBU 2023–25; **Organized** National Science Day 2024, GBU